

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration: 1 Hour
Total Marks:50

Annexure A

Scenario Based Assignment

Code of Conduct:

I SUBISH A G certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Subish A G

Annexure A

Scenario Based Assignment

1. Scenario : A government deploys AI-powered drones to deliver emergency medicines in a flood-affected region. The environment is highly dynamic—roads may suddenly become inaccessible, and weather conditions affect travel cost. The drone has access to: A heuristic estimate (straight-line distance), Partial real-time updates about blocked paths and Variable movement costs depending on terrain and weather. However, due to urgency, the drone must quickly decide routes with minimum delay and risk.

Tasks:

- i. Explain how Greedy Best-First Search would behave in this scenario and discuss its strengths and weaknesses under uncertainty
- ii. Explain how A* would handle the same situation, including how it balances cost and heuristic
- iii. Analyze the impact of heuristic accuracy on both algorithms
- iv. Discuss how dynamic changes (blocked paths) affect search performance
- v. Recommend the most suitable algorithm and justify your decision considering real-time constraints, optimality, and safety

2. Scenario: A smart city implements an AI system to optimize traffic signal timings across hundreds of intersections. The objective is to minimize Total waiting time, Fuel consumption and Traffic congestion. The system must continuously adjust signals based on traffic flow. However the search space is extremely large, traffic patterns change dynamically and the system may get stuck in locally optimal solutions

Tasks. Formulate this as an optimization problem and explain why traditional search is inefficient

- i. Explain how Hill Climbing would work in this scenario and identify its limitations
- ii. Discuss issues like local maxima, plateaus, and ridges with real-world interpretation
- iii. Explain how Simulated Annealing or Genetic Algorithms can overcome these limitations
- iv. Recommend the best approach and justify your choice based on scalability and adaptability

3. Scenario: An autonomous rover is deployed on Mars to explore unknown terrain. It must: navigate safely to collect samples, avoid hazards (craters, rocks), operate with limited sensing capability and make decisions without a complete map. The rover updates its knowledge as it explores, making this a real-time decision-making problem.

Tasks:

- i. Explain why this problem requires an online search agent instead of offline search
- ii. Analyze the challenges of partial observability and dynamic environments

- iii. Describe how the rover builds and updates its internal model
- iv. Compare online search with uninformed and informed search strategies
- v. Justify the most suitable approach considering uncertainty, adaptability, and safety

4. **Scenario:** A university is designing an AI system to automatically generate exam timetables. The system must satisfy multiple constraints: No student should have overlapping exams, Limited number of exam halls, Certain subjects must be scheduled before others, Faculty availability constraints, Special accommodations for some students, The complexity increases as the number of courses and students grows.

Tasks :

Formulate the problem as a Constraint Satisfaction Problem (CSP)

- i. Clearly define variables, domains, and constraints with explanation
- ii. Explain how backtracking search works in this scenario
- iii. Discuss how constraint propagation (e.g., forward checking) improves efficiency
- iv. Analyze real-world challenges and justify the best strategy for scalability

5. **Scenario:** Strategic Game AI for Real-Time Decision Making (Minimax & Alpha-Beta Pruning) A gaming company is developing an AI opponent for a real-time strategy game. The AI must compete against human players, Make decisions within strict time limits, Evaluate complex game states with multiple possible moves, The decision tree is extremely large, making computation expensive.

Tasks:

- i. Explain how the Minimax algorithm models this problem
- ii. Analyze the computational challenges of large game trees
- iii. Explain how Alpha-Beta pruning improves efficiency with detailed reasoning
- iv. Design an evaluation function and justify the factors considered
- v. Recommend a strategy for balancing optimality and real-time performance

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

1. Ans: Introduction

A government deploys AI-powered drones to deliver emergency medicines in flood-affected areas. The drone operates in a dynamic environment where roads may become blocked unexpectedly, weather changes movement costs, and only partial information is available. The AI must quickly determine a safe and efficient route.

(i) Greedy Best-First Search (GBFS)

Working

Greedy Best-First Search selects the next node with the lowest heuristic value $h(n)$.

It ignores the actual cost already travelled.

Evaluation function:

$$\begin{aligned} &[\\ f(n) &= h(n) \\ &] \end{aligned}$$

where

- $h(n)$ = Straight-line distance to destination.

Example

Drone always chooses the location that appears closest to the hospital.

Strengths

- Very fast
- Low memory usage
- Suitable when urgent decisions are needed
- Works well if heuristic is accurate

Weaknesses

- Ignores travel cost
- Cannot guarantee shortest route
- May choose dangerous paths
- Performs poorly when blocked roads appear suddenly
- Can revisit unnecessary nodes

(ii) A* Search

Working

A* combines

- Actual travel cost

- Estimated remaining cost

Evaluation function

[

$$f(n)=g(n)+h(n)$$

]

where

$g(n)$ =Actual travel cost

$h(n)$ =Estimated remaining distance

Example

The drone considers

- Weather
- Terrain
- Energy consumption
- Distance

before selecting the next route.

Advantages

- Finds optimal path
- Balances speed and cost
- Handles varying movement costs
- Avoids expensive routes

Disadvantages

- More computation
- Requires additional memory
- Needs heuristic calculation

(iii) Impact of Heuristic Accuracy

Highly Accurate Heuristic

GBFS

- Very fast
- Near-optimal path

A*

- Fast
- Optimal path
- Minimal node expansion

Poor Heuristic

GBFS

- Wrong decisions
- Dead ends
- Longer travel

A*

- Still finds optimal path
- Searches more nodes
- Slower execution

(iv) Effect of Dynamic Blocked Paths

Flood conditions continuously change.

Challenges

- Roads become inaccessible
- Weather increases travel cost
- Partial information

GBFS

- May continue toward blocked roads
- Needs complete replanning

A*

- Updates path cost
- Recalculates shortest path
- More robust

(v) Recommended Algorithm

Recommended: A*

Justification

Criteria	Greedy BFS A*	
Speed	High	Moderate
Safety	Medium	High
Optimality	No	Yes
Dynamic Cost	Poor	Excellent

Criteria	Greedy BFS	A*
Reliability	Medium	High

Although Greedy BFS is faster, emergency medicine delivery prioritizes patient safety over a small increase in computation. A* provides safer and more reliable routes while considering terrain and weather.

2. Ans:

Objective Function

Minimize

- Total waiting time
- Fuel consumption
- Traffic congestion

Decision Variables

- Green signal duration
- Signal order
- Offset timing

Constraints

- Traffic demand
- Emergency vehicle priority
- Maximum cycle time

Why Traditional Search is Inefficient

There are hundreds of intersections.

Each signal has multiple timing options.

Search space becomes

[
 n^m
]

where

n = timing options

m = intersections

Traditional search becomes computationally impossible.

(i) Hill Climbing

Hill Climbing continuously improves the current solution.

Steps

1. Start with initial timing.
2. Evaluate traffic performance.
3. Move to better neighboring solution.
4. Repeat.

Advantages

- Simple
- Fast
- Low memory

Limitations

- Gets stuck at local optimum
- Cannot backtrack
- Sensitive to starting point

(ii) Local Maxima, Plateau, Ridge

Local Maximum

Traffic improves locally but not globally.

Example

One junction improves while neighboring intersections worsen.

Plateau

Many timing plans give nearly identical performance.

AI cannot determine the better direction.

Ridge

Best solution requires multiple simultaneous timing changes.

Hill Climbing changes only one variable.

(iii) Simulated Annealing and Genetic Algorithm

Simulated Annealing

Allows occasional worse solutions.

Advantages

- Escapes local maxima
- Better global optimization

Genetic Algorithm

Population of solutions

Operations

- Selection
- Crossover
- Mutation

Advantages

- Explores many solutions
- Handles large search spaces
- Adapts to changing traffic

(iv) Recommendation

Best Algorithm: Genetic Algorithm

Reasons

- Highly scalable
- Parallel optimization
- Adapts continuously
- Avoids local optimum
- Suitable for smart cities

3. Ans:

(i) Why Online Search?

Offline search requires complete map.

Mars rover has

- Unknown terrain
- Limited sensing
- Dynamic hazards

Therefore, online search is necessary.

(ii) Challenges

Partial Observability

Cannot see entire terrain.

Dynamic Environment

New obstacles appear.

Limited Resources

Battery

Communication delay

Sensor noise

(iii) Internal Model

The rover continuously updates

- Safe locations
- Hazard locations
- Sample locations
- Battery status

This knowledge improves future decisions.

(iv) Comparison

Feature	Uninformed	Informed	Online
Complete Map	Yes	Yes	No
Heuristic	No	Yes	Optional
Unknown Environment	Poor	Moderate	Excellent
Adaptability	Low	Medium	High

(v) Recommendation

Online Search Agent

Reasons

- Learns while moving
- Handles uncertainty
- Safe navigation
- Real-time decisions

4. Ans:

(i) Variables, Domains and Constraints

Variables

Each course exam.

Domains

Available day, time, room.

Constraints

- Student conflict
- Faculty conflict
- Hall limit
- Accessibility
- Subject ordering

(ii) Backtracking Search

Steps

Assign first exam.

Check constraints.

If conflict occurs

Backtrack.

Assign another slot.

Repeat.

(iii) Constraint Propagation

Forward Checking

Removes invalid future assignments immediately.

Benefits

- Fewer conflicts
- Faster search
- Reduced computation

(iv) Recommendation

Backtracking + Forward Checking + Constraint Propagation

Reasons

- Efficient
- Scalable

- Reduces search space
- Produces valid timetable

Scenario 5: Game AI (Minimax and Alpha-Beta Pruning)

(i) Minimax

Players

MAX = AI

MIN = Human

The AI selects moves that maximize its score while assuming the opponent minimizes it.

(ii) Computational Challenges

Large branching factor

Huge search tree

High computation

Memory intensive

Real-time limits

(iii) Alpha-Beta Pruning

Idea

Skip branches that cannot influence the final decision.

Benefits

- Fewer node evaluations
- Faster decisions
- Same optimal result as Minimax

(iv) Evaluation Function

Example

[

Score=(5\times Resources)+(10\times Army)+(20\times Territory)-(8\times EnemyPower)

]

Factors

- Resources
- Army strength
- Territory control

- Enemy strength
- Health
- Remaining time

These factors reflect the overall strategic advantage of the AI.

(v) Recommendation

Best Approach

Alpha-Beta Pruning with Minimax

Reasons

- Same optimal decision as Minimax
- Much faster search
- Suitable for real-time games
- Handles large decision trees efficiently